

A.R.I.E.S.

AUTOMATED ROBOTIC INDUSTRIAL EQUIPMENT SYSTEM

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BRIEFLY.

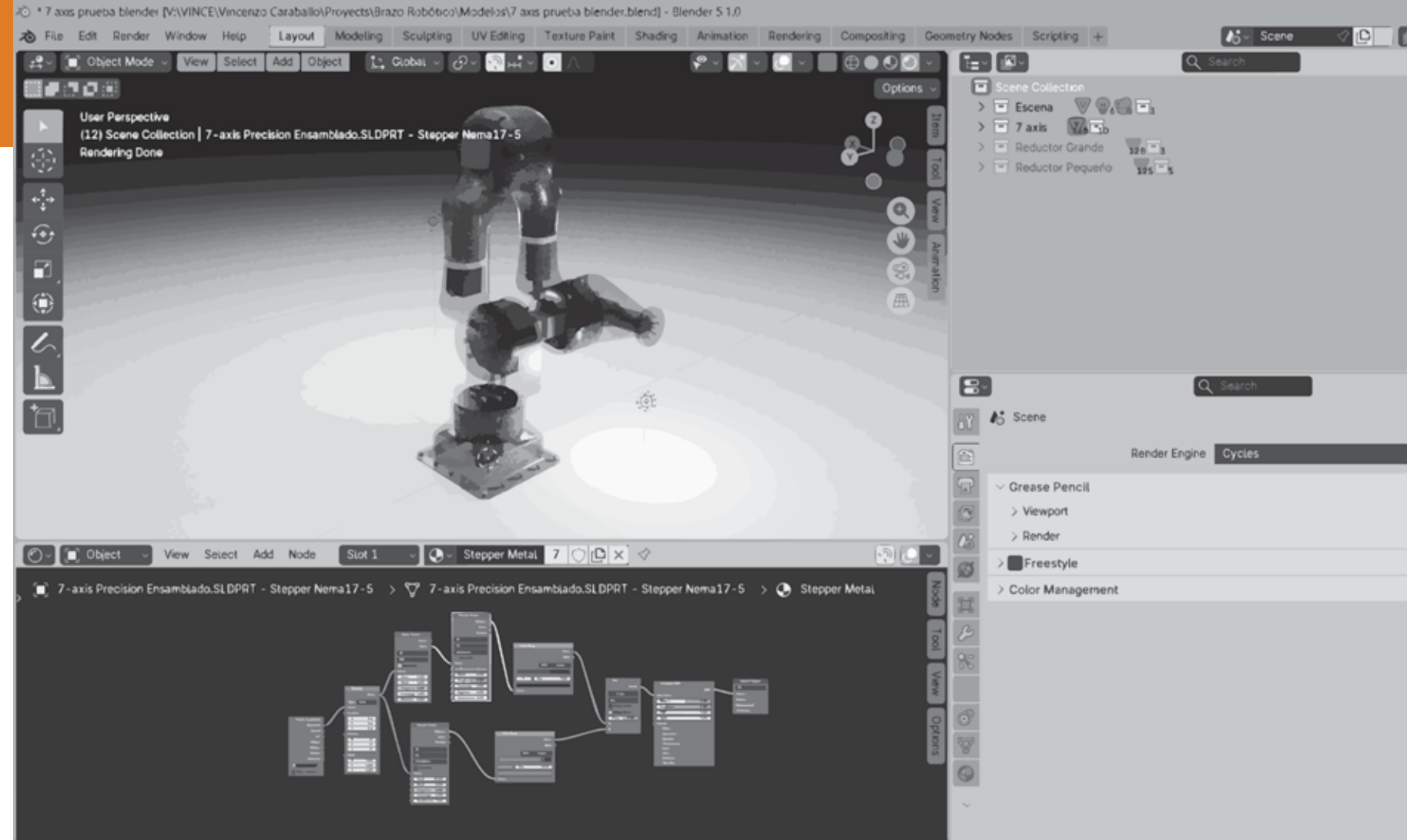
AUTOMATION SYSTEMS FOR MANUFACTURING

THE AIM OF THE INITIATIVE IS TO ENABLE SMALL AND MEDIUM-SIZED MEXICAN COMPANIES TO IMPLEMENT AUTOMATION SOLUTIONS IN THEIR CONSTRUCTION AND OPERATIONAL PROCESSES, OPTIMISING PRODUCTION, ENABLING THE INTEGRATION OF ADVANCED PROCESSES AND, BROADLY SPEAKING, IMPROVING THE COMPETITIVENESS OF MEXICO'S INDUSTRIAL SECTOR ON THE INTERNATIONAL STAGE.

THIS PROJECT PROPOSES THE DEVELOPMENT OF PRODUCTS, MACHINES AND SOFTWARE WHICH, TOGETHER, FORM A COMPETITIVE AND ACCESSIBLE ECOSYSTEM FOR THE CURRENT MARKET. TO ACHIEVE THIS, THERE ARE THREE ESSENTIAL PILLARS:

- PRECISE LINEAR AND ANGULAR ACTUATORS THAT ENABLE CNC (COMPUTERISED NUMERICAL CONTROL) SYSTEMS TO TRANSLATE VIRTUAL DYNAMICS OR PROCESSES INTO THE PHYSICAL REALM.
- EQUIPMENT THAT ENABLES THE USE OF THESE ACTUATORS AS COORDINATED UNITS FOR THE MANUFACTURE, ASSEMBLY, ORGANISATION AND TRANSPORT OF PRODUCTS
- A PLATFORM THAT, IN AN INTUITIVE AND EFFICIENT MANNER, ENABLES THE PROGRAMMING AND OPERATION OF INDIVIDUAL MACHINES AND UNIT ASSEMBLIES FOR THE AUTOMATION OF A COMPANY'S SPECIFIC PROCESSES.

AS AN ADDITIONAL COMPETITIVE ADVANTAGE, WE PROPOSE TO INVESTIGATE THE POSSIBILITY OF HYBRIDLY INTEGRATING TRADITIONAL MANUFACTURING TECHNIQUES WITH ADDITIVE MANUFACTURING FOR THE PRODUCTION OF PARTS, MOULDS AND ACCESSORIES FOR PRODUCTS AND MACHINES, ACHIEVING FORMAL SOPHISTICATION, HIGHER PRODUCTION VOLUMES AND A WIDE RANGE OF FORMAL PROPERTIES, DEMONSTRATING THE VALUE OF THESE TECHNOLOGIES IN THE MODERN INDUSTRIAL LANDSCAPE.



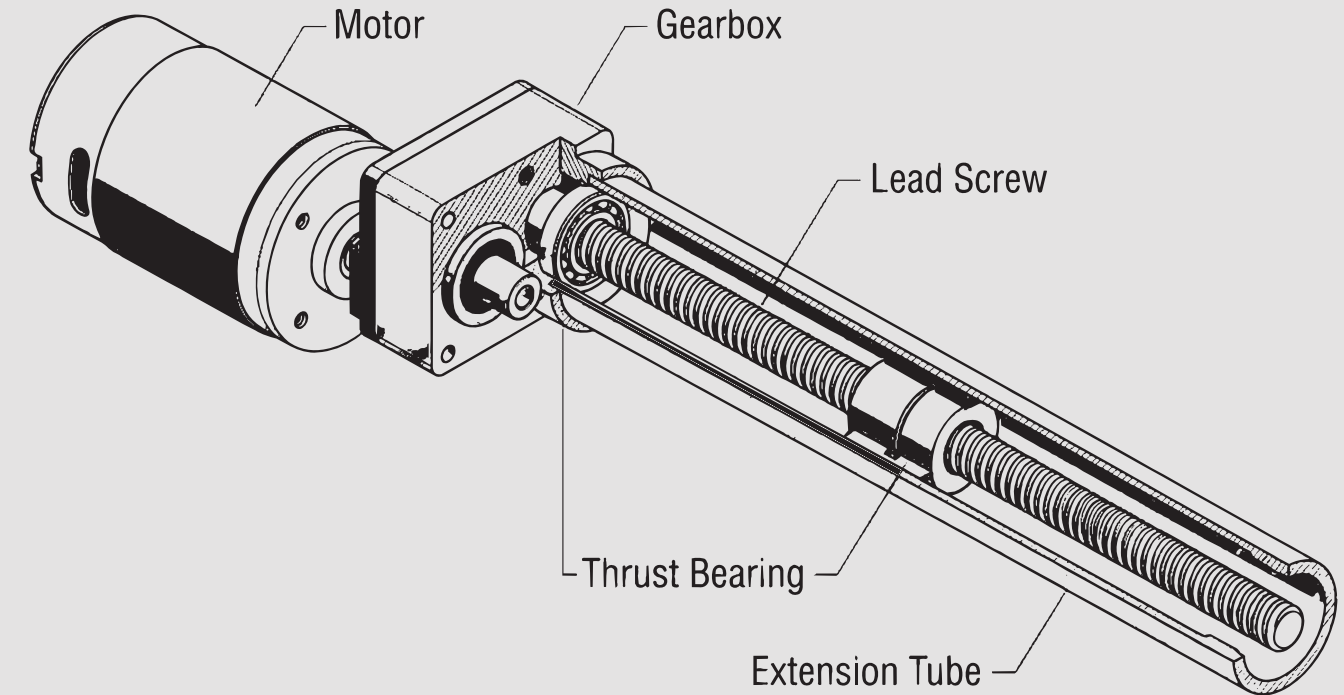
1. THE ACTUATOR.

PRECISION TRANSMISSIONS

THE FIRST COMPONENT IN ANY MECHANICAL SYSTEM IS AN ACTUATOR: REFERRING TO ANY COMPONENT THAT ENABLES THE CONTROLLED MOVEMENT OF AN OBJECT IN SPACE, ACTUATORS ARE MAINLY DIVIDED INTO LINEAR AND ANGULAR TYPES, DEPENDING ON THE TYPE OF MOVEMENT THEY PRODUCE.

TO THIS END, ACTUATORS COMPRISE SEVERAL ELEMENTS THAT ENABLE THEIR FUNCTION:

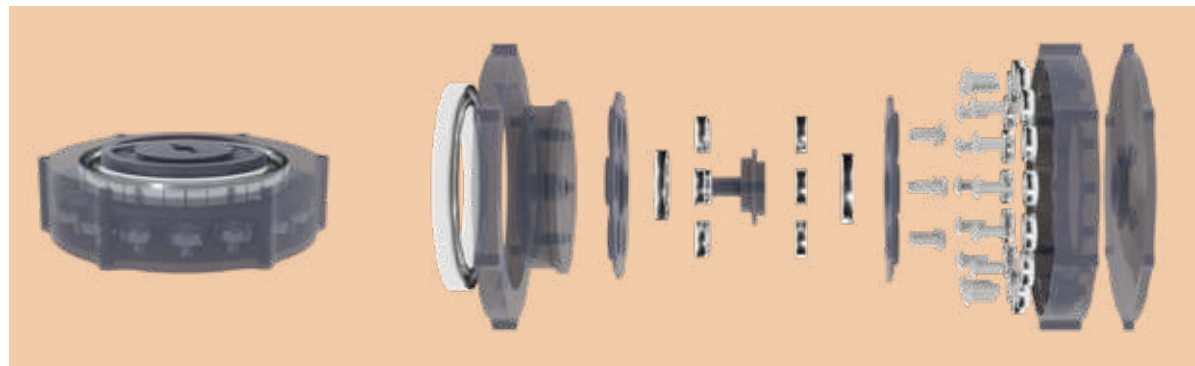
- THE MOTOR, OR PISTON, WHICH GENERATES MOVEMENT BY UTILISING ELECTRICAL ENERGY
- COMPRESSED AIR OR ANY EXTERNAL POTENTIAL FORCE
- THE TRANSMISSION, WHICH REDIRECTS, CONVERTS OR REFINES THE MOTOR'S MOVEMENT TO TRANSLATE IT INTO A VIABLE OUTPUT
- THE ENCODER, WHICH ACTS AS A TRANSLATOR AND CONTROLLER FOR THE MOTOR, ENABLING THE MEASUREMENT, OPERATION AND REGULATION OF ITS MOVEMENT
- THE GUIDE ALONG WHICH THE FINAL MOVEMENT TAKES PLACE, TYPICALLY BY MOVING A FINAL EFFECTOR OR



CYCLOIDAL REDUCER.

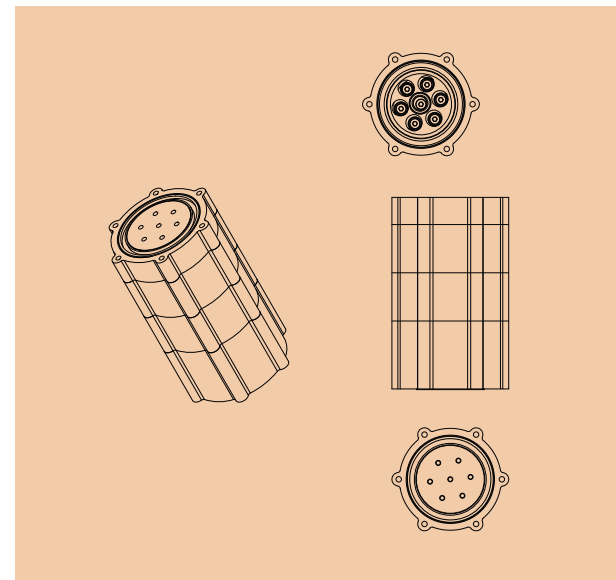
MODULAR DESIGN OF MULTIPLICATIVE TRANSMISSION

THE FIRST PRODUCT OF THE PROJECT IS A COMPONENT OF THE ACTUATORS: THE GEARBOX OR REDUCTOR. IN THIS CASE, THE PROPOSAL IS TO DESIGN A FAMILY OF REDUCTOR MODULES WHICH, BY COMBINING DIFFERENT MODELS, ENABLE THE CREATION OF CUSTOMISED CYCLOIDAL TRANSMISSIONS WITH VARIABLE REDUCTION RATIOS, PROVIDING HIGH-RESOLUTION MOTION, POWER TRANSMISSION AND BACKLASH-FREE LOCKING FOR ALL MECHANICAL MOTION PROCESSES.



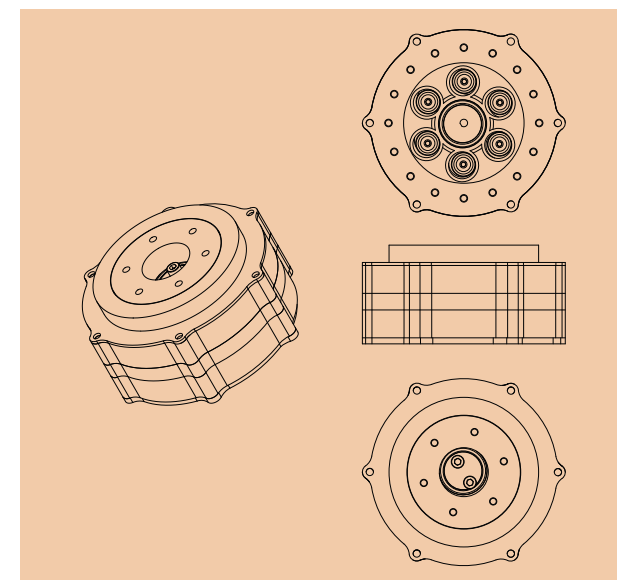
REDUCTION MODULE 225:1

TO EXPAND THE RANGE OF APPLICATIONS, TWO ALTERNATIVE TWO-STAGE TRANSMISSION PRODUCTS HAVE BEEN DEVELOPED, COMBINING CNC MACHINING PROCESSES WITH 3D ADDITIVE MANUFACTURED POLYMER PARTS (FDM, SLS, RESIN), PROVIDING A BALANCE OF QUALITY, DURABILITY, COST AND VIABILITY FOR MASS PRODUCTION.



TUBULAR REDUCER 100:1

- RATIO: 18.75mm
- HEIGHT: 70.75mm
- REDUCER MODULE: 10:1
- STAGES: 2 SUPERPOSED STAGES
- INPUT AXIS: 5mm FOR NEMA17



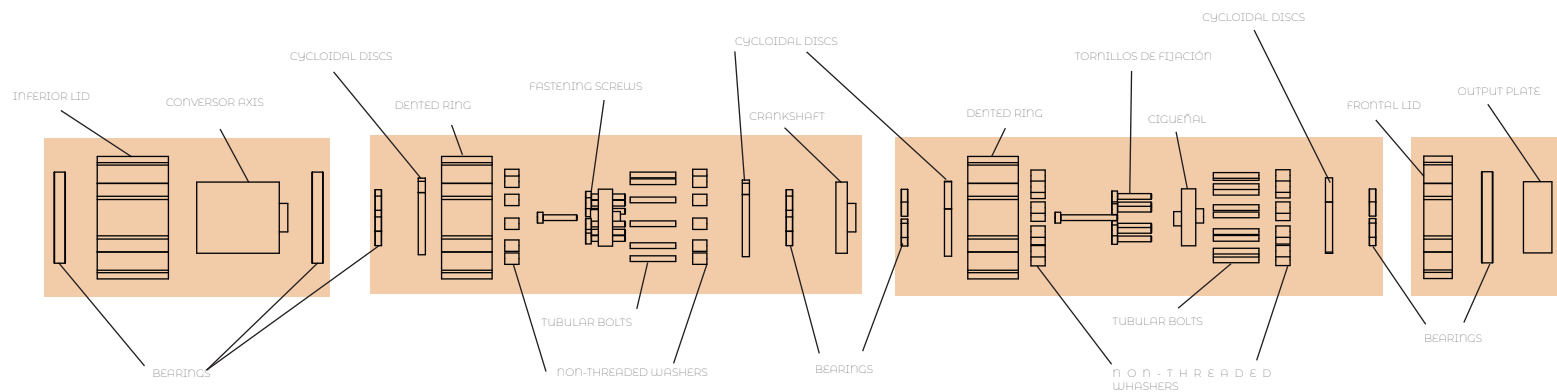
COMPACT REDUCER 225:1

- RATIO: 61mm
- HEIGHT: 65.5mm
- REDUCER MODULE: 15:1
- STAGES: 2 SUPERPOSED STAGES
- INPUT AXIS: 5mm FOR NEMA17

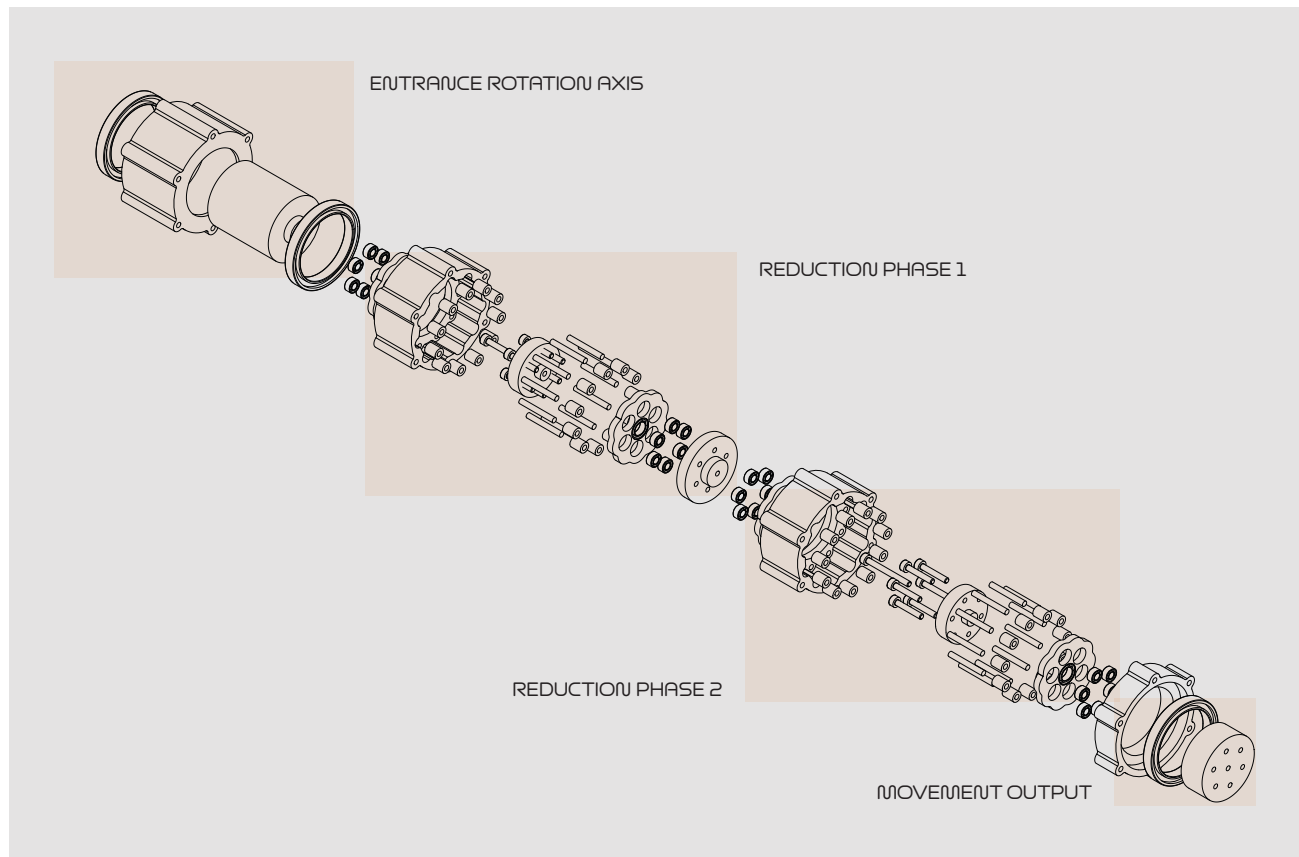
THE INTERNAL MECHANISM.

EPICYCLOIDAL REDUCER BASED IN ANTI-PHASE MOVEMENT

THE REDUCTION GEARBOX IS BASED ON THE TRANSLATIONAL MOTION GENERATED BY THE ROTATIONAL OSCILLATION OF EPICYCLOID DISCS AS THEY ROTATE IN CONSTANT CONTACT WITH A TOOTHED RING ON WHICH THE DISCS SLIDE, MAINTAINING CONTACT BETWEEN SURFACES, THIS ALLOWS THE TRANSLATION OF SECONDARY AXES ON THE DISC WITH A DIFFERENCE FROM THE CENTRAL AXIS, GENERATING AN OUTPUT OF REDUCED-SPEED MOTION, MULTIPLIED FORCE AND VIRTUALLY ZERO backlash, DUE TO THE ABSOLUTE CONTACT BETWEEN PARTS.



EXPLOITED VIEW AND MECHANISM SECTIONS

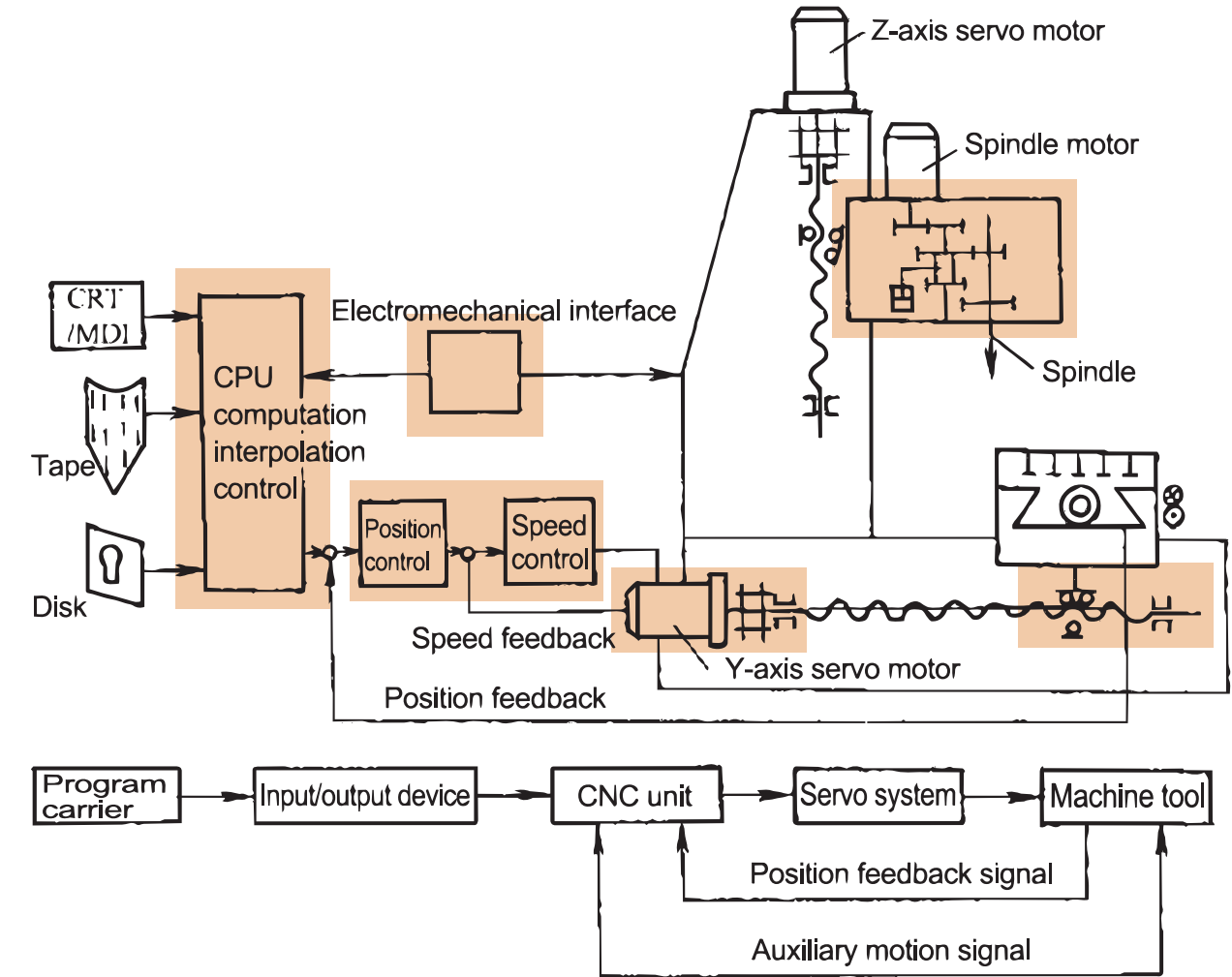


2. THE UNIT.

CNC AUTOMATED UNITS

THE SECOND ELEMENT OF THE PROJECT IS CNC EQUIPMENT. DEVELOPED IN MASSACHUSETTS IN 1958, AND POPULARISED WORLDWIDE AS AN ESSENTIAL PART OF INDUSTRIAL MECHATRONICS, COMPUTERISED NUMERICAL CONTROL TECHNOLOGY, OR 'CNC', REFERS TO THE PROCESS OF TRANSLATING DIGITAL COMMANDS BASED ON VECTORIAL PATHWAYS DEFINED BY COORDINATES TO PRODUCE MOVEMENT, USING LINEAR OR ANGULAR ACTUATORS WHICH, TOGETHER, ENABLE THE PROGRAMMING, OPERATION AND AUTOMATION OF MOVEMENTS, SEQUENCES AND COMPLETE PROCESSES BY INTRODUCING TOOLS, MANIPULATORS OR SENSORS TO THE MOBILE TERMINALS OF THIS EQUIPMENT. CNC UNITS ESSENTIALLY CONSIST OF 6 MAIN COMPONENTS:

- THE FRAME: THE STRUCTURE ON WHICH THE EQUIPMENT IS BUILT, MEETING THE PHYSICAL REQUIREMENTS AND ACCOMMODATING THE COMPONENTS NECESSARY FOR ITS OPERATION, WHETHER A SINGLE PIECE OR AN ASSEMBLY.
- THE ACTUATORS: LINEAR OR ANGULAR, THEY ARE RESPONSIBLE FOR MOVEMENT
- THE ENCODERS: THE TRANSDUCER THAT ENABLES COMMUNICATION BETWEEN MOTORS AND
- CONTROLLER: THE BRAIN OF THE UNIT, RESPONSIBLE FOR RECEIVING INSTRUCTIONS, COMMANDS OR CONDITIONS, AS WELL AS RECEIVING FEEDBACK FROM PHYSICAL SENSORS TO COORDINATE AND DIRECT MOVEMENT, CARRYING OUT THE PROGRAMMED TASKS.
- THE SENSORS: COMPONENTS THAT ENABLE FEEDBACK TO BE RECEIVED REGARDING THE ENVIRONMENT AND THE MOVEMENT OF THE EQUIPMENT, TO INFORM THE CONTROLLER.
- THE EFFECTOR: THE FINAL SECTION OF THE ROBOT, WHERE THE POWER CONNECTIONS AND THE ROBOT'S TOOLS ARE LOCATED. IT IS THE CENTRAL POINT OF THE ROBOT, RESPONSIBLE FOR PERFORMING THE ROBOT'S TASKS



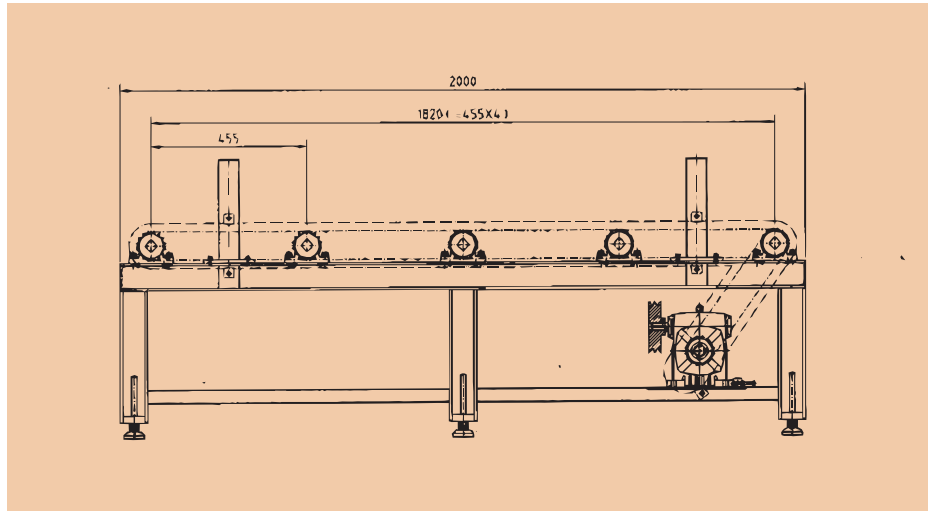
MECHANICAL DEVICES.

MACHINE CLASSIFICATION

COMPUTATIONAL NUMERICAL CONTROL EQUIPMENT CAN BE CLASSIFIED IN TWO MAIN WAYS: BY COMPLEXITY AND BY AUTONOMY. THE CLASSIFICATION BY COMPLEXITY IS EXPLAINED BELOW, BUT INDEPENDENT OF THIS, CNC EQUIPMENT OR SYSTEMS ARE DIVIDED ACCORDING TO THE SAE J3016 CLASSIFICATION: LEVEL 0 (MANUAL OPERATION), LEVEL 1 (ASSISTED OPERATION), LEVEL 2 (PARTIAL AUTOMATION), LEVEL 3 (CONDITIONAL AUTOMATION), LEVEL 4 (HIGH INDIVIDUAL AUTOMATION) AND LEVEL 5 (FULL-AUTOMATION ECOSYSTEM)

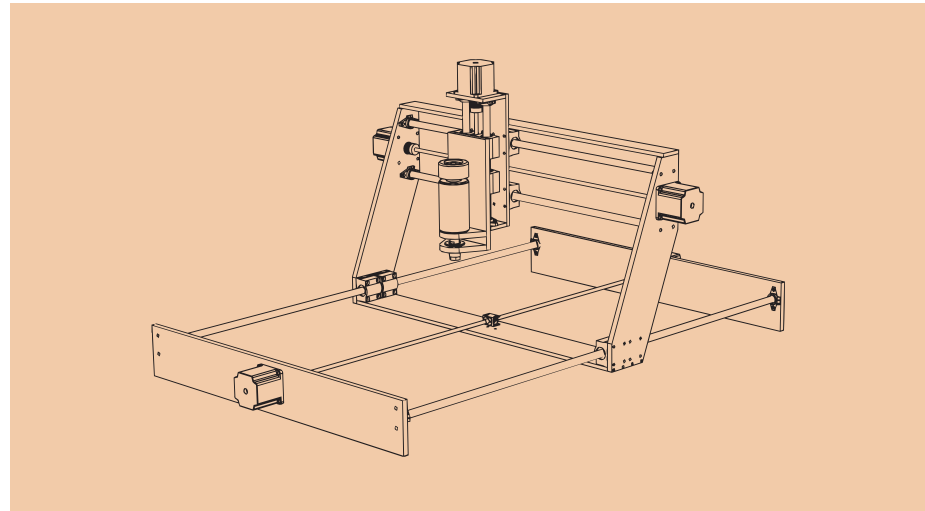
SIMPLE MACHINES

- 1-2 DEGREES OF FREEDOM
- MOVEMENT OF PARTS OR COMPONENTS
- REPETITIVE AND HIGH-SPEED TASKS
- PRECISION AT THE EXPENSE OF COMPLEXITY



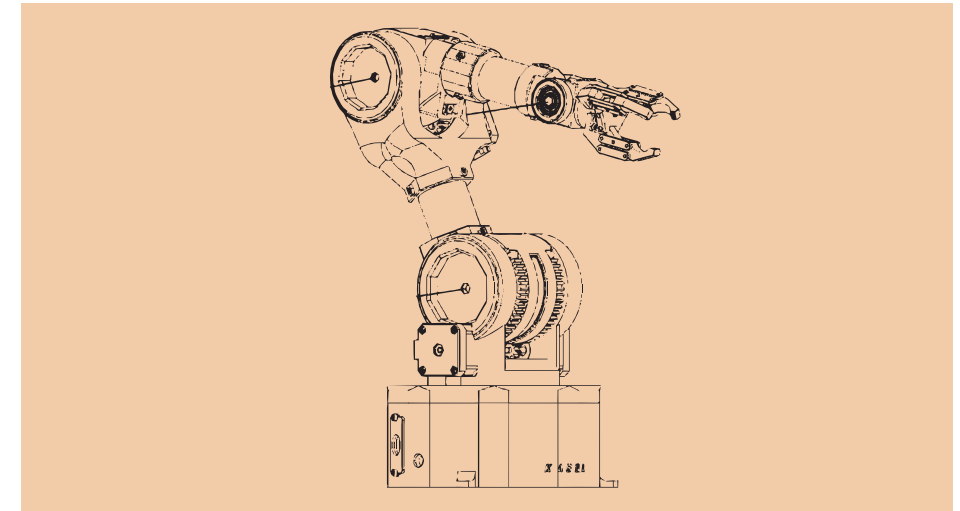
CMM (COORDINATE MEASURING MACHINE)

- COMBINATION OF LINEAR MOTIONS
- VECTORIAL PATHS BASED ON CARTESIAN COORDINATES
- 2-5 DEGREES OF FREEDOM (3 LINEAR, 2 ROTATIONAL)
- RANGE OF MOTION DEFINED BY THE STRUCTURAL FRAME



COMPLEX MACHINES

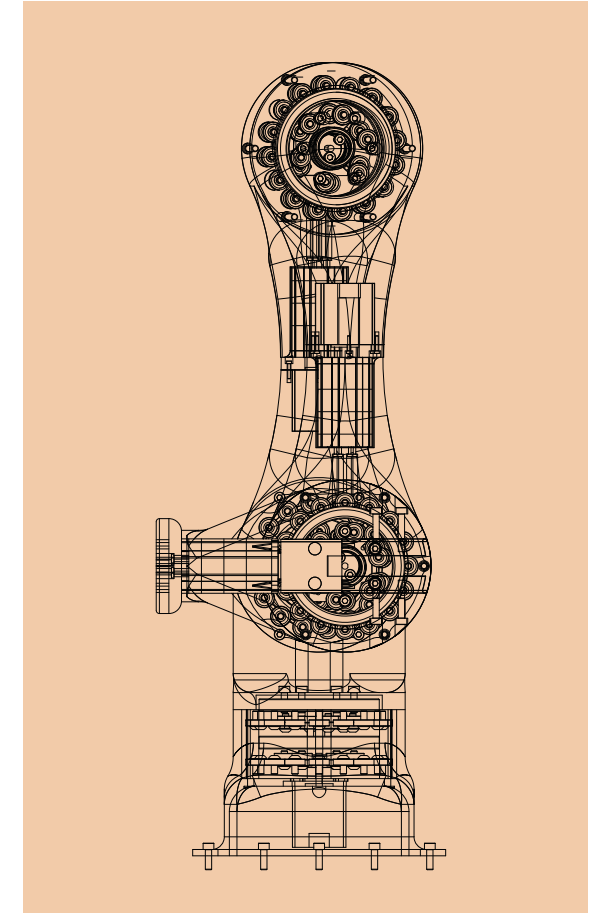
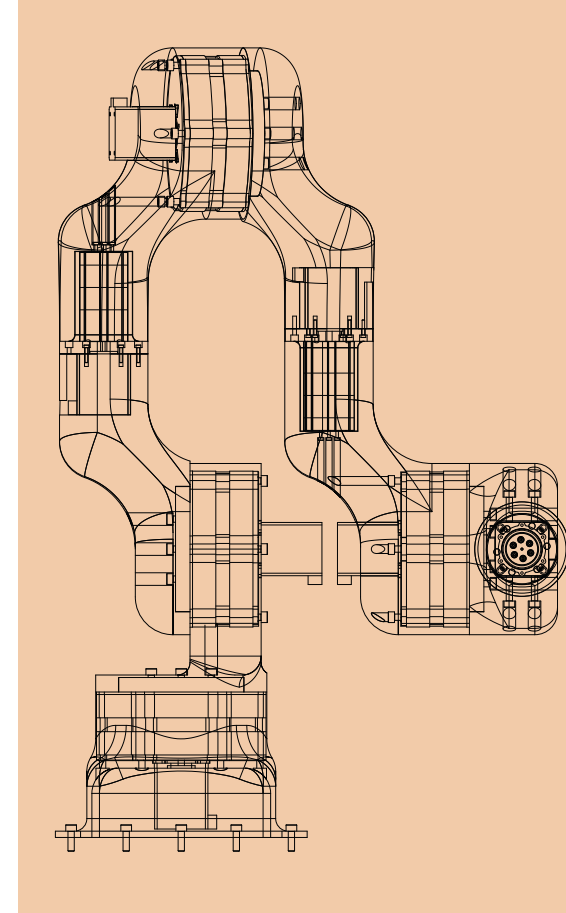
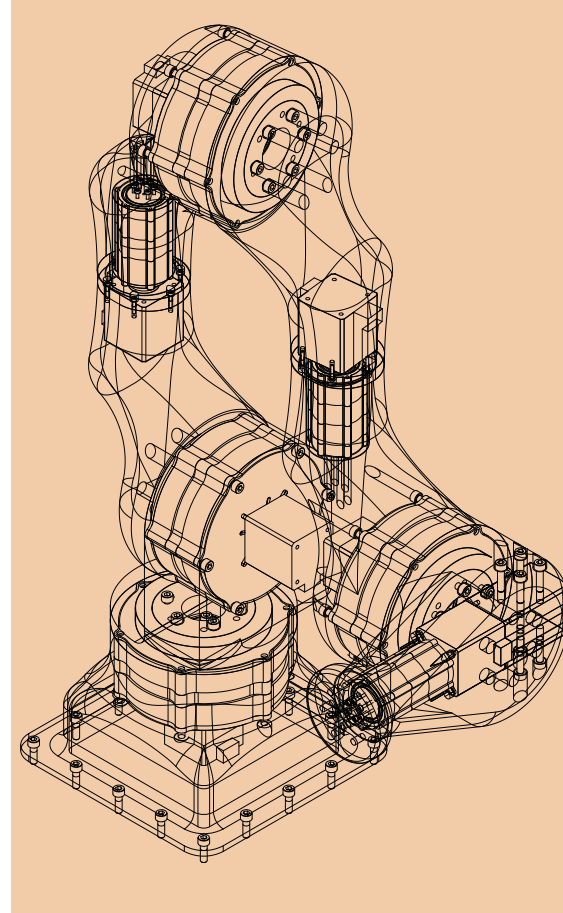
- 3 TO 8 DEGREES OF MOTION
- LINEAR AND ANGULAR FREEDOM OF MOVEMENT
- COMPLEX TASKS AND COORDINATION BETWEEN UNITS
- EXPANDABLE RANGE OF ACTION WITH A SLIDING BASE

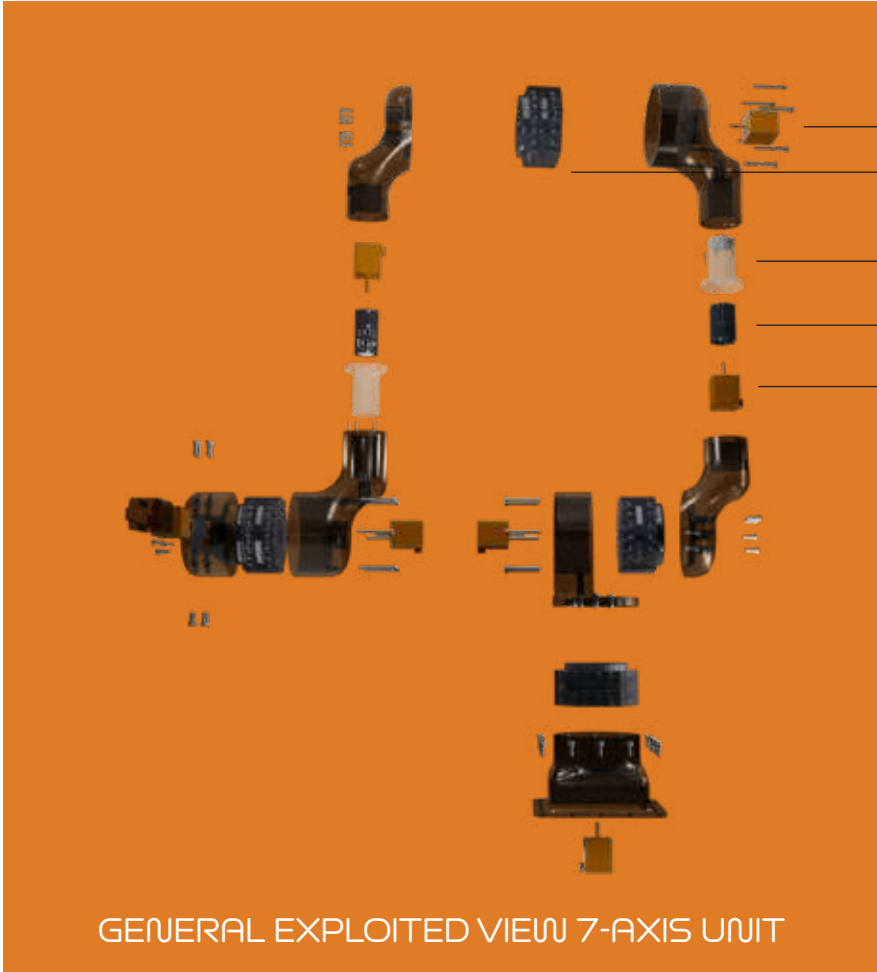


THE UNIT.

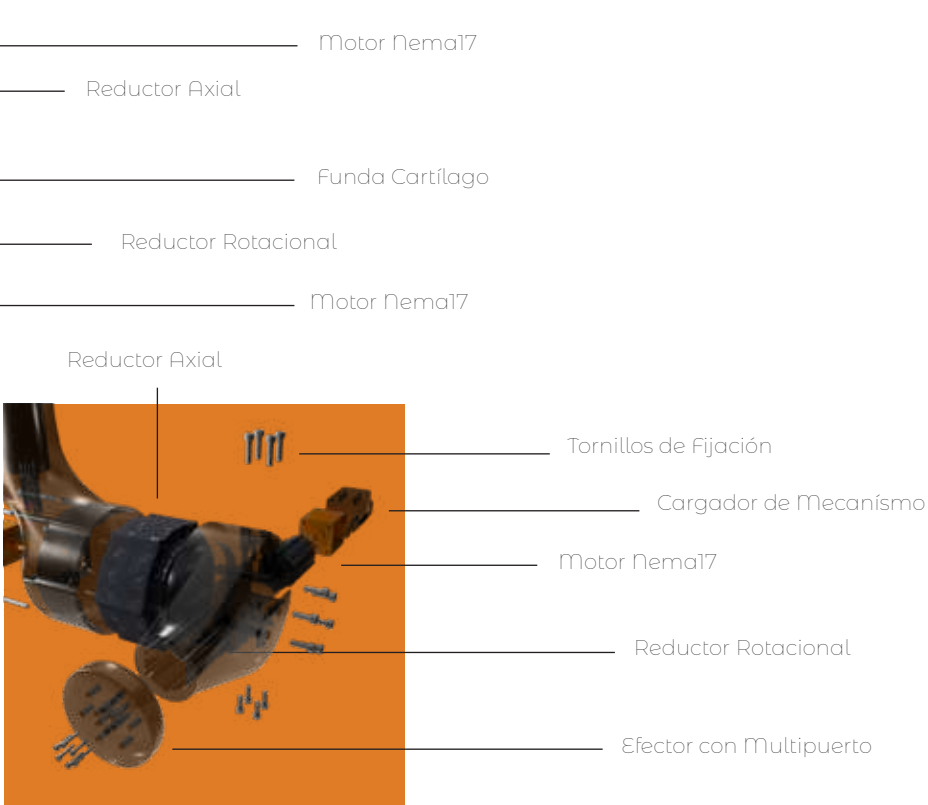
FLEXIBLE-MANUFACTURE CELLS

WITHIN THREE-DIMENSIONAL COORDINATE VECTOR CONTROL SYSTEMS, THE GOLD STANDARD IS THE ROBOTIC ARM: THE ULTIMATE MACHINE FOR AXIAL AND ANGULAR CONTROL, USED IN THE MANUFACTURING AND ASSEMBLY LINES OF COMPANIES SUCH AS BMW AND MERCEDES-BENZ. IT FEATURES 3 SEGMENTS OF ANGULAR MOVEMENT AND 4 ROTATIONAL JOINTS, ALLOWING A 300° RANGE OF MOTION IN ALL DIRECTIONS, WITH VARIABLE EXTENSION CAPACITY AND THE ABILITY TO USE VARIOUS TOOLS THANKS TO AN INDEPENDENTLY MOUNTED END-EFFECTOR. FOR THESE REASONS, COMBINED WITH ITS VERSATILITY OF USE, COMPATIBILITY BETWEEN UNITS AND COLLABORATIVE WORKING WITH OTHER PRODUCTION, ORGANISATION AND ASSEMBLY MACHINES, THE IMPLEMENTATION OF THIS TYPE OF UNIT, IN CONJUNCTION WITH DISTRIBUTION LINES AND LIFTING PLATFORMS AT AFFORDABLE COSTS, COULD ENABLE THE FULL AUTOMATION OF STORAGE WAREHOUSES, ASSEMBLY LINES AND PRODUCTION FACILITIES, MINIMISING HUMAN ERROR, EXTENDING OPERATING HOURS AND OPTIMISING INDUSTRIAL ACTIVITY.





GENERAL EXPLOITED VIEW 7-AXIS UNIT



EXPLOITED FOCUS VIEW



TOP-VIEW



FRONT VIEW



SIDE VIEW

ARIES: ROBOTIC DEVICE.

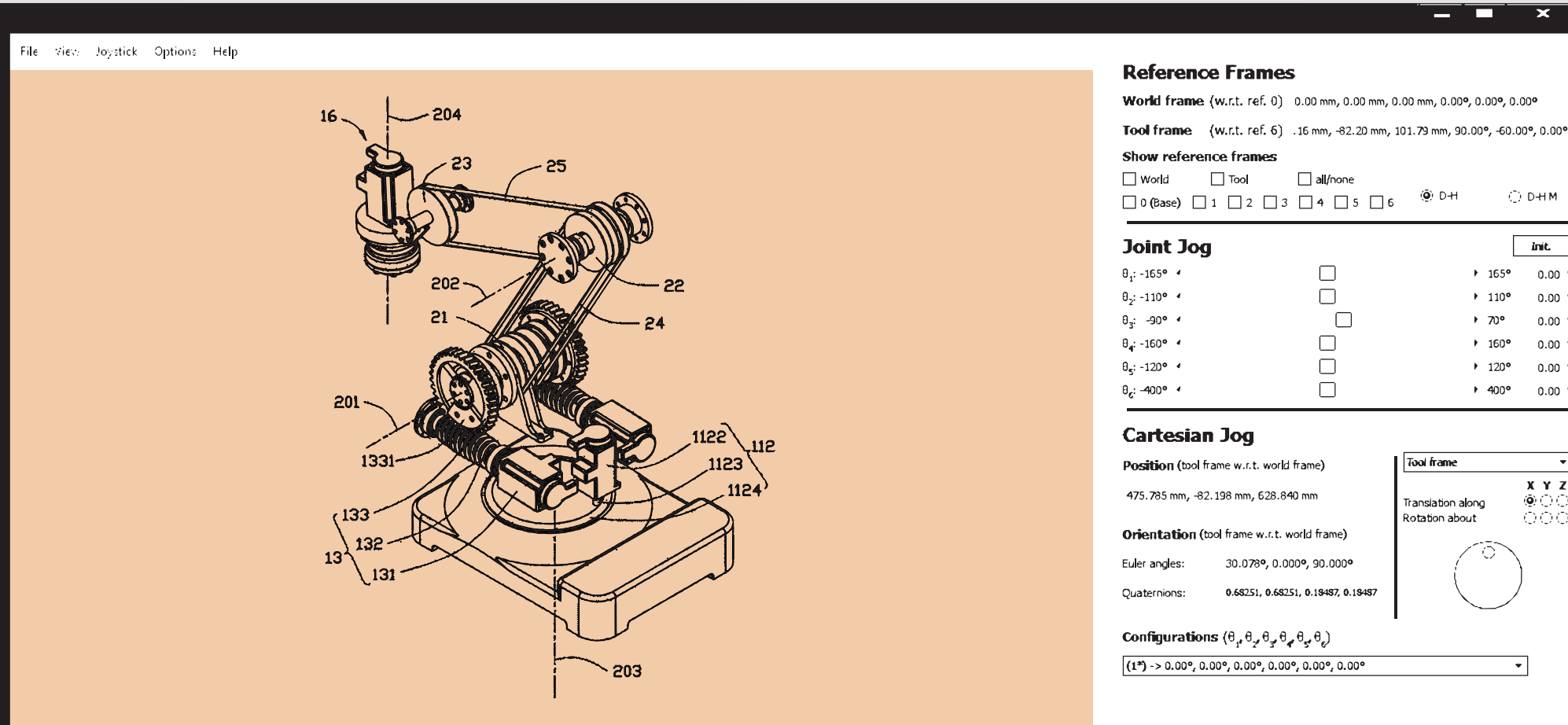
3. THE PROGRAM.

KINEMATICS AND MOVEMENT STUDY, COORDINATION AND OPERATION

THE THIRD COMPONENT CAN BE UNDERSTOOD AS A COMBINATION OF VARIOUS FACTORS:

- THE DIGITAL SIMULATOR OF GEOMETRIES, MECHANISMS AND KINEMATICS.
- THE PROGRAMMING AND INSTRUCTION INTERFACE FOR TRAINING AND GENERATING INSTRUCTIONS
- THE SLICER OR TRAJECTORY GENERATOR FOR CONVERTING VECTORS INTO COORDINATED INSTRUCTIONS
- THE RECEIVER ALGORITHM WHICH ENABLES, THROUGH THE PHYSICAL COMPONENTS, THE EXECUTION OF PROGRAMMED INSTRUCTIONS AND/OR MANUAL OPERATION OF THE EQUIPMENT.

HOWEVER, WHILE THE ABOVE-MENTIONED ELEMENTS MAY BE FOUND IN EXISTING SOFTWARE, WHETHER FREE, PAID OR OPEN-SOURCE, THE REAL OBJECTIVE OF THE THIRD COMPONENT IS TO ENABLE THE INTERACTION AND COOPERATION OF EACH INDIVIDUAL COMPONENT AS AN ASSEMBLY LINE THROUGH ADD-ONS THAT ACT AS LINKS, UNITING THE TOOLS INTO AN ACCESSIBLE, CUSTOMISABLE AND VERSATILE ECOSYSTEM FOR THE PROGRAMMING AND OPERATION OF UNITS OR CELLS.



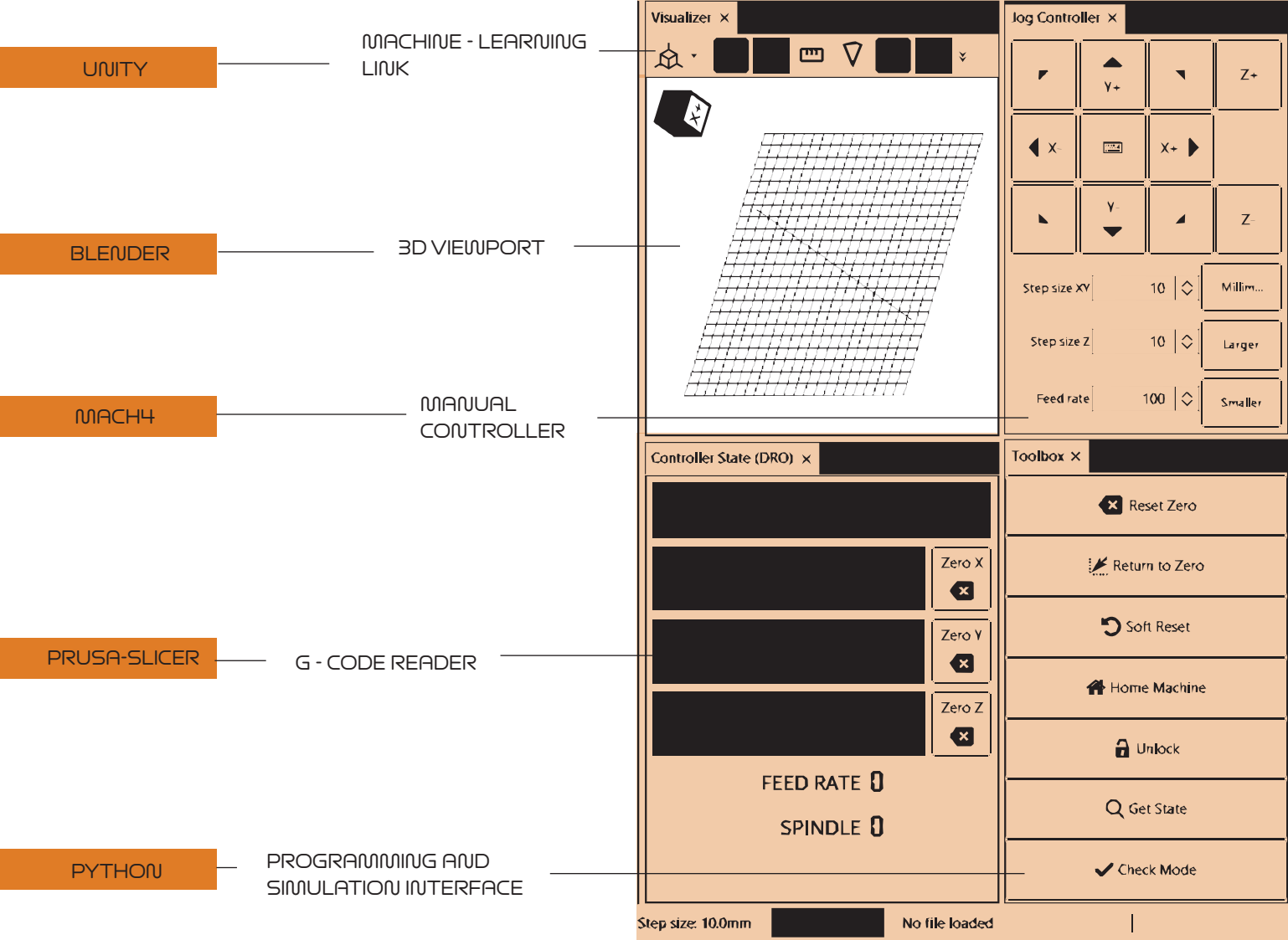
PLATFORM AND PLUG-INS.

INTER-PLATFORM ECOSYSTEM

IN SHORT, THE PROGRAM CAN BE SUMMARISED AS A SERIES OF ADD-ONS OR ‘PLUG-INS’ WHICH, USING AN OPEN-SOURCE 3D DESIGN AND PROGRAMMING PLATFORM AS A BASIS, ENABLE THE CREATION OF LINKS BETWEEN SPECIALISED PLATFORMS, CENTRALISED WITHIN A COHESIVE WORKSPACE, EXPLOITING THE SPECIALISED ASPECTS OF EACH COMPONENT TO CONTRIBUTE TO DEFINED STAGES OF THE PROCESS, WHILE ALLOWING A DEGREE OF FLEXIBILITY AND CUSTOMISATION THAT ADAPTS TO THE USER’S AUTOMATION NEEDS. THE USE OF OPEN-SOURCE PLATFORMS ALLOWS FOR THE INTERCONNECTION OF RESOURCES AND FREE EXPERIMENTATION, WHILE OPENING THE DOORS TO REPOSITORIES OF USEFUL INFORMATION, MODIFICATIONS AND EXPANSIONS CREATED BY THE DIGITAL COMMUNITY, WHICH, HAVING FREE ACCESS TO THESE TOOLS, RESOLVES ISSUES AND SHARES THESE SOLUTIONS WITH THE PUBLIC THROUGH PROJECTS, THUS FACILITATING SPECIALISED WORK.

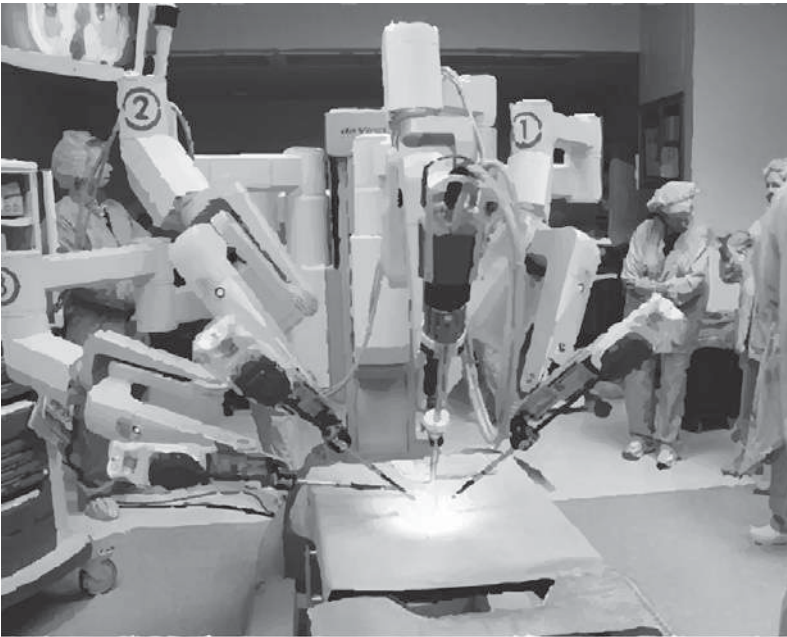
TO ASSEMBLE THIS PLATFORM, THE FOLLOWING COMPONENTS HAVE BEEN OUTLINED:

- BLENDER: THE FOUNDATION OF THE PROGRAMME, A 3D MODELLING, SIMULATION, ANIMATION, PROGRAMMING AND RENDERING PROGRAMME CAPABLE OF CREATING AND INTEGRATING PYTHON-BASED PLUGINS
- PYTHON/MATLAB: A MATHEMATICAL PROGRAMMING AND SIMULATION INTERFACE FOR CREATING PROGRAMMES OR ‘SCRIPTS’ SPECIALISED IN DIGITAL TOOLS, 3D STUDIES, MACHINE LEARNING AND OBJECT-ORIENTED PROGRAMMING.
- PRUSA-SLICER: SOFTWARE FOR PROCESSING AND TRANSLATING DIGITAL GEOMETRIES AND CREATING COORDINATE-BASED INSTRUCTIONS OR “G-CODE” SPECIALISED IN THE CALIBRATION AND OPERATION OF CNC MACHINES.
- UNITY: 3D DESIGN AND PROGRAMMING ENGINE, FOCUSED ON THE SIMULATION OF ANIMATIONS, MECHANICAL STUDIES AND TRAINING PROCESSES OR “MACHINE LEARNING” OF COORDINATED UNITS AND CELLS, ROUTE OPTIMISATION AND
- MACH4: INTERFACE FOR CONNECTION, RECOGNITION, CALIBRATION, MANUAL OPERATION AND LINKING TO PROGRAMMING AND AUTOMATION PLATFORMS FOR CNC EQUIPMENT.



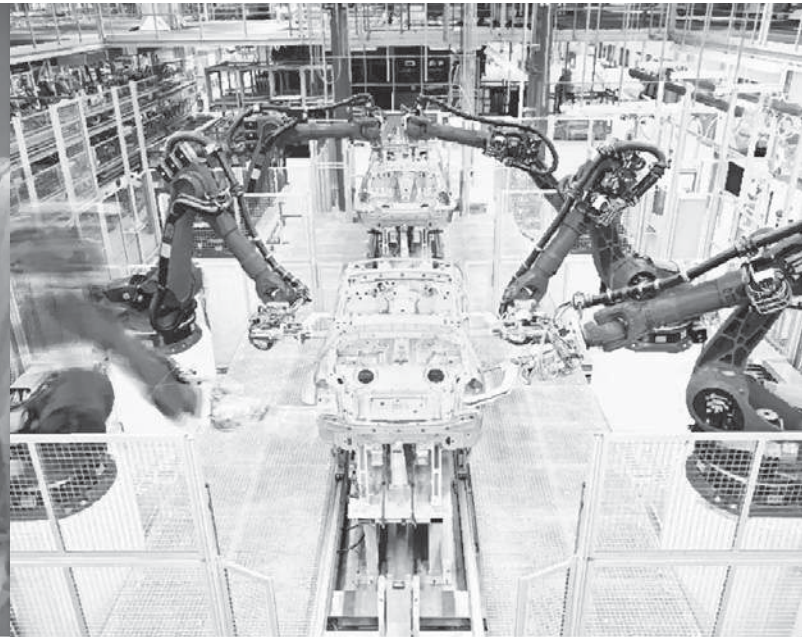
USE CASES.

MEDIC



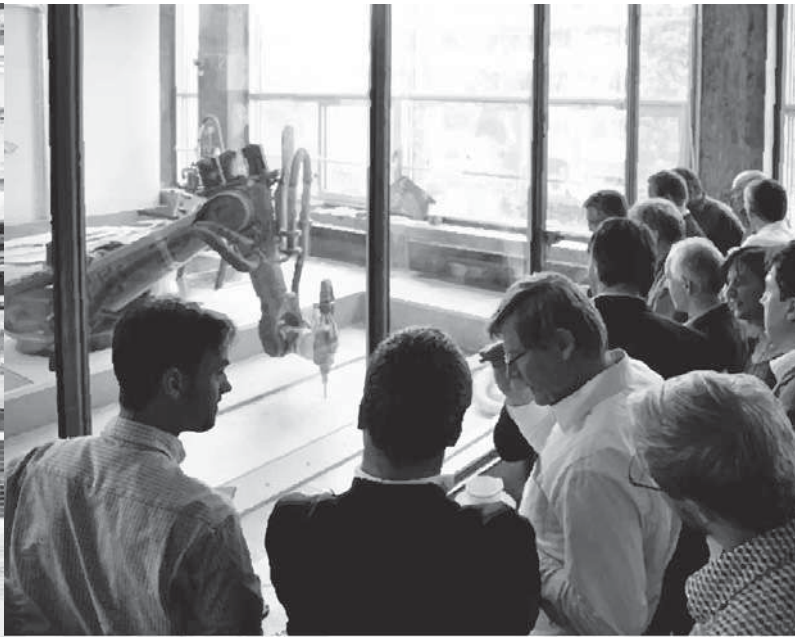
- OPERATIONS AND PROFESSIONAL SUPPORT
- TOOL PREPARATION AND MAINTENANCE
- QUALITY CONTROL
- MANUFACTURING AND PRODUCTION

INDUSTRIAL



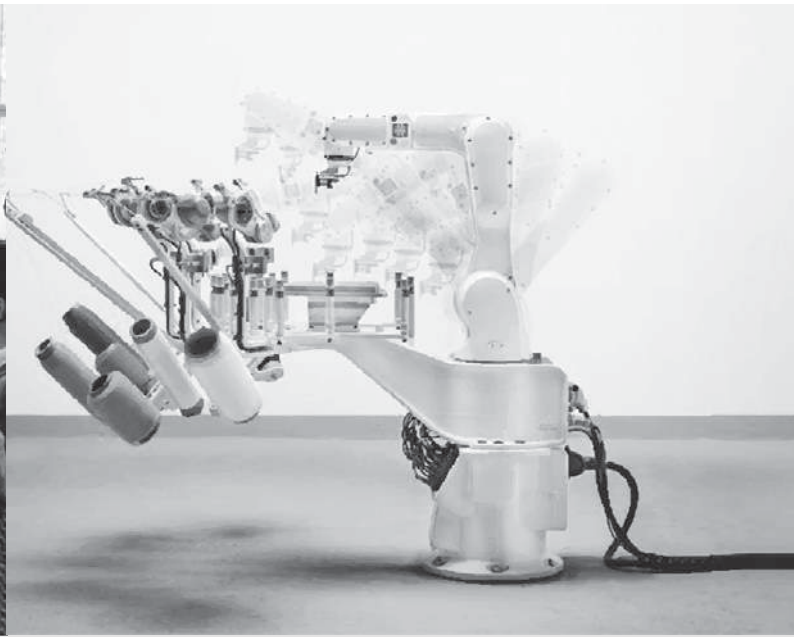
- MANUFACTURING
- ASSEMBLY
- SORTING AND LOGISTICS
- QUALITY CONTROL
- TRANSPORT

ACADEMIA



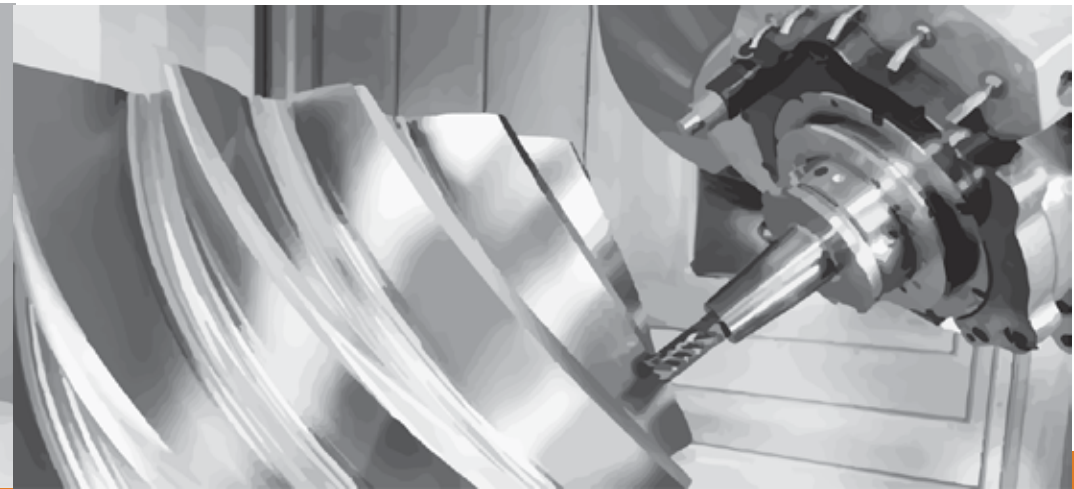
- INDUSTRIAL ENGINEERING
- BUSINESS ADMINISTRATION
- ROBOTICS AND PROGRAMMING
- DESIGN AND ARCHITECTURE
- SERVICES AND MANAGEMENT

RESEARCH



- ADDITIVE AND SUBTRACTIVE MANUFACTURING
- AUTONOMOUS WORK CELLS
- ASSISTANCE AND ACCESSIBILITY
- REGENERATIVE PRODUCTION
- LOW-IMPACT SYSTEMS

MATERIAL.



EQUIPMENT:

- HAAS ALUMINIUM AND METAL MACHINING STATION
- HIGH-RESOLUTION SLS OR RESIN PRINTER (8-16K)
- MICROMETRIC CALIBRATION EQUIPMENT (MITUTOYO)
- VOLTMETER, TORQUE METER AND BAUMANOMETER
- PCB DESIGN AND MANUFACTURING STATION
- SOLDERING AND FINISHING KIT

RESOURCES:

- COMPUTING EQUIPMENT FOR SIMULATION AND PROGRAMMING
- 3D SOFTWARE AND PROGRAMMING LICENCES
- DIGITAL MANUFACTURING AND SIMULATION ACCESSORIES
- PRECISION FASTENERS (TIMKEN)
- RAW MATERIALS FOR MANUFACTURING (RESIN, STEEL, ETC.)
- MOTION AND ACTUATION COMPONENTS (STEPPERONLINE)
- VARIOUS EFFECTORS (VACUUM PUMP, EXTRUDER, GRIPPER, ETC.)
- MISCELLANEOUS (LUBRICATING OILS AND GREASES, VALVES, ETC.)

SERVICES:

- MACHINING OF VARIOUS MATERIALS
- OBJECT-ORIENTED PROGRAMMING
- MACHINE LEARNING AND DEEP LEARNING SIMULATION
- MECHATRONICS CONSULTANCY
- RENT OR SUBSIDY FOR WORKSPACE

CURRENT STATE.

1. THE ACTUATOR:

- TECHNOLOGICAL READINESS LEVEL: TRL2
- LOW-FIDELITY FUNCTIONAL PROTOTYPE
- DESIGN STUDIES AND VARIOUS MODELS
- MODULAR PRODUCT FAMILY
- MANUFACTURING COST ANALYSIS

2. EL UNIT:

- TECHNOLOGICAL READINESS LEVEL: TRL1
- DIGITAL MODEL AND PHYSICAL SIMULATIONS
- INITIAL STAGES OF KINEMATICS STUDY
- QUOTATIONS FROM VARIOUS MANUFACTURERS
- STUDY OF COMMERCIAL ALTERNATIVES

3. THE PROGRAM:

- TECHNOLOGICAL MATURITY LEVEL: CONCEPT PHASE
- SELECTION OF PROGRAMMES FOR THE ECOSYSTEM
- COORDINATED MOVEMENT PILOT PROGRAMME
- BASIC RESOURCES FOR KINEMATICS AND MACHINE-L



PROJECT PHASES.

PHASE 1: REDUCER

- MODELS
- PROTOTYPE
- COSTS
- SERVICES
- MATERIALS
- BEARINGS
- FASTENERS
- ELECTRONICS
- CALIBRATION
- REPEATABILITY
- BENCHMARK
- USAGE TEST

PHASE 2: UNIT

- MODELS
- PROTOTYPE
- COSTS
- SERVICES
- MATERIALS
- BEARINGS
- FASTENERS
- ELECTRONICS
- SYSTEMS
- SENSORS
- CALIBRATION
- REPEATABILITY
- TOOLS
- COORDINATION
- PARALLELISM
- FUNCTIONAL TESTING

PHASE 3: PROGRAM

- SKELETON
- KINEMATICS
- ANIMATION
- VECTORS
- COORDINATES
- LABELLING
- MOTION
- MACHINE-L
- SIMULATION
- EXTRAPOLATION
- PROGRAMMING
- PARALLELISM
- TESTING
- INSERTION
- COORDINATION
- INTEGRATION

PHASE 4: CELL TEAMS

- DESIGNATION
- UNIT TYPES
- MODELS
- SIMULATION
- COORDINATION
- REFINEMENT
- MACHINE-L
- TOOLS
- PROCESSES
- INDEPENDENCE
- FEEDBACK
- REINTEGRATION
- ITERATION
- PHYSICAL TESTING
- HYBRIDISATION





THANK YOU.

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